Aliter: You may also use the method of detached coefficients:

$$\begin{array}{rrrr} & 2 & +3 & -6 \\ & & +4 & -5 \\ \hline & -10 & -15 & +30 \\ 8 & +12 & -24 & \\ \hline 8 & +2 & -39 & +30 \end{array}$$

$\therefore \quad (4x-5)(2x^2+3x-6) = 8x^3 + 2x^2 - 39x + 30$

Exercise 3.1

1. Which of the following expressions are polynomials. If not give reason:

   (i) $\frac{1}{x^2} + 3x - 4$ (ii) $x^2(x-1)$

   (iii) $\frac{1}{x}(x+5)$ (iv) $\frac{1}{x^{-2}} + \frac{1}{x^{-1}} + 7$

   (v) $\sqrt{5}\,x^2 + \sqrt{3}\,x + \sqrt{2}$ (vi) $m^2 - \sqrt[3]{m} + 7m - 10$

2. Write the coefficient of x^2 and x in each of the following polynomials.

   (i) $4 + \frac{2}{5}x^2 - 3x$ (ii) $6 - 2x^2 + 3x^3 - \sqrt{7}\,x$

   (iii) $\pi x^2 - x + 2$ (iv) $\sqrt{3}\,x^2 + \sqrt{2}\,x + 0.5$

   (v) $x^2 - \frac{7}{2}x + 8$

3. Find the degree of the following polynomials.

   (i) $1 - \sqrt{2}\,y^2 + y^7$ (ii) $\frac{x^3 - x^4 + 6x^6}{x^2}$

   (iii) $x^3(x^2 + x)$ (iv) $3x^4 + 9x^2 + 27x^6$

   (v) $2\sqrt{5}\,p^4 - \frac{8p^3}{\sqrt{3}} + \frac{2p^2}{7}$

4. Rewrite the following polynomial in standard form.

   (i) $x - 9 + \sqrt{7}\,x^3 + 6x^2$ (ii) $\sqrt{2}\,x^2 - \frac{7}{2}x^4 + x - 5x^3$

   (iii) $7x^3 - \frac{6}{5}x^2 + 4x - 1$ (iv) $y^2 + \sqrt{5}\,y^3 - 11 - \frac{7}{3}y + 9y^4$

5. Add the following polynomials and find the degree of the resultant polynomial.

   (i) $p(x) = 6x^2 - 7x + 2$ $q(x) = 6x^3 - 7x + 15$

   (ii) $h(x) = 7x^3 - 6x + 1$ $f(x) = 7x^2 + 17x - 9$

   (iii) $f(x) = 16x^4 - 5x^2 + 9$ $g(x) = -6x^3 + 7x - 15$

6. Subtract the second polynomial from the first polynomial and find the degree of the resultant polynomial.

   (i) $p(x) = 7x^2 + 6x - 1$ $q(x) = 6x - 9$

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